Plastic Raw Material

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
HDPE Resin (Blow Moulding Grade); HDPE Resin (Injection Moulding Grade)	IS 7328 (High Density Polyethylene (HDPE) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	HDPE Resin (Injection Moulding Grade);ISO 17855-1 (Plastics — Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics by Density-Gradient Technique); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate of Thermoplastics); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 527 (Tensile Properties of Plastics)	FDA 21 CFR 177 (Food Contact Grades); UL 94 (where flame classification applies); Plastics Industry Resin Classification Guidelines
HDPE Resin (Pipe Grade PE80); HDPE Resin (Pipe Grade PE100)	IS 7328 (HDPE Material); IS 4984 (HDPE Pipes for Water Supply);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	HDPE Resin (Pipe Grade PE100);ISO 12162 (Thermoplastics Materials for Pipes and Fittings for Pressure Applications — Classification and Designation); ISO 17855-1 (Polyethylene (PE) Moulding and Extrusion Materials)	ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D2837 (Hydrostatic Design Basis for Thermoplastic Pipe Materials); ASTM D1505 (Density of Plastics); ASTM D1238 (Melt Flow Rates of Thermoplastics)	DIN EN ISO 9080 (Extrapolation of Hydrostatic Strength of Thermoplastics Pipes); DIN EN ISO 12162 (Thermoplastics Materials for Pipes and Fittings); DIN EN ISO 1872-1 (Polyethylene Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	PPI (Plastics Pipe Institute) Technical Guidelines; AWWA Plastic Pipe Standards; CSA B137 Series (Pressure Piping Applications)
HDPE Resin (Film Grade)	IS 7328;IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 17855-1 (Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D882 (Tensile Properties of Thin Plastic Sheeting)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density); DIN EN ISO 527 (Tensile Properties)	FDA 21 CFR 177 (Food Contact Applications); Flexible Packaging Association Guidelines; European Food Contact Requirements
HDPE Resin (Monofilament Grade)	IS 7328;IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 17855-1 (Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D1505 (Density of Plastics)	DIN EN ISO 1872-1 (Polyethylene Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density)	FDA 21 CFR 177 (Food Contact Applications, where applicable); Textile Industry Fibre Specifications
LDPE Resin (Film Grade); LDPE Resin (Coating Grade); LLDPE Resin (Stretch Film Grade)	IS 2508 (Polyethylene Materials for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 14636 (Flexible Packaging	LDPE Resin (Coating Grade); LLDPE Resin (Stretch Film Grade);ISO 17855-1 (Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D882 (Tensile Properties of Thin Plastic Sheeting)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 527-1 (Tensile Properties)	FDA 21 CFR 177; Flexible Packaging Association Guidelines; Packaging Industry Association Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	for Food Products) (Film Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360				
LDPE Resin (Wire & Cable Grade)	IS 7328;IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 1872-1 (Plastics — Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D1248 (Polyethylene Plastics Extrusion Materials for Wire and Cable); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D1525 (Vicat Softening Temperature of Plastics)	DIN EN IEC 60811-501 (Mechanical Tests for Insulation and Sheathing Materials of Electrical Cables); DIN EN ISO 1872-1 (Polyethylene Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate)	UL 1581 (Wire and Cable Testing); UL 44 (Thermoset-Insulated Wires and Cables, where applicable); National Electrical Manufacturers Association Requirements
LLDPE Resin (Rotomoulding Grade)	IS 7328;IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 17855-1 (Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics)	DIN EN ISO 1872-1 (Polyethylene Moulding Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 1183-1 (Density); DIN EN ISO 527 (Tensile Properties)	Rotational Moulding Industry Association (ARM) Guidelines; FDA 21 CFR 177 (Food Contact Applications)
MDPE Resin (Gas Pipe Grade)	IS 7328;IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 12162 (Thermoplastics Materials for Pipes and Fittings for Pressure Applications — Classification and Designation); ISO 17855-1 (Polyethylene Materials)	ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D2837 (Hydrostatic Design Basis for Thermoplastic Pipe Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics)	DIN EN ISO 1872-1 (Polyethylene Materials); DIN EN 1555-1 (Plastics Piping Systems for Gas Supply); DIN EN ISO 12162 (Thermoplastics Materials for Pipes and Fittings); DIN EN ISO 9080 (Hydrostatic Strength)	AGA Gas Distribution Standards; PPI Technical Guidelines; PHMSA Pipeline Safety Requirements
PP Homopolymer (Injection Grade); PP Homopolymer (Film Grade); PP Random Copolymer	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10146; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	PP Homopolymer (Film Grade); PP Random Copolymer;ISO 19069-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 177; UL 94 (Electrical Applications); Plastics Industry Material Classification Guidelines
PP Homopolymer (Raffia Grade); PP Block Copolymer	IS 10910 (Polypropylene and its Copolymers); IS 16187 (PP Woven Sacks) (Raffia Applications); IS 7328; IS 10146; IS	PP Block Copolymer;ISO 19069-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 1873-1 (Polypropylene Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 179-1 (Charpy Impact Strength); DIN EN ISO 527 (Tensile Properties)	Packaging Industry Association Guidelines; FDA 21 CFR 177 (Food Contact Applications); FIBCA Guidelines (where woven sack applications apply)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360				
PP-R Resin	IS 15801;IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 15874-1 (Plastics Piping Systems for Hot and Cold Water Installations — Polypropylene (PP) — General); ISO 19069-1 (Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM F2389 (Pressure-rated Polypropylene (PP) Piping Systems); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics)	DIN EN ISO 15874-1 (Polypropylene Piping Systems for Hot and Cold Water); DIN EN ISO 1873-1 (Polypropylene Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties)	NSF/ANSI 61 (Drinking Water Components); NSF/ANSI 14 (Plastic Piping Systems); CSA B137 Series
PP-H Resin	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10146; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 19069-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	Chemical Process Industry Specifications; ASME B31.3 (Process Piping Applications); NSF/ANSI 61 (where applicable)
PP-B Resin	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10146; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 19069-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength); DIN EN ISO 178 (Flexural Properties)	Plastic Drainage Industry Specifications; NSF/ANSI 14; Municipal Piping Authority Requirements
PVC Suspension Resin; PVC Emulsion Resin	IS 10151 (Polyvinyl Chloride (PVC) and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10146; IS 10910; IS 12252; IS 11434; IS 14534; IS 2530; IS 13360	PVC Emulsion Resin;ISO 1628-2 (Plastics — Determination of Viscosity of Polymers in Dilute Solution — Polyvinyl Chloride Resins); ISO 1060-1 (Plastics — Poly(vinyl chloride) — Designation System and Basis for Specifications)	ASTM D1755 (Poly(Vinyl Chloride) Resins); ASTM D1243 (Intrinsic Viscosity of Poly(Vinyl Chloride)); ASTM D1894 (Coefficient of Friction of Plastic Film and Sheeting, where applicable); ASTM D3367 (Poly(Vinyl Chloride) Resins Used in Plastics)	DIN EN ISO 1060-1 (PVC Resins — Determination of K-Value and Viscosity Number); DIN EN ISO 1628-2 (Viscosity of Polymers — PVC Resins); DIN EN ISO 21306-1 (PVC-U Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties, where applicable)	UL 94 (where flame classification applies); Vinyl Institute Guidelines; Chemical Industry Resin Specifications
CPVC Resin	IS 15778;IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 11613 (Plastics — Chlorinated Poly(vinyl chloride) (PVC-C) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D1784 (Rigid Poly(Vinyl Chloride) and Chlorinated Poly(Vinyl Chloride) Compounds); ASTM D2122 (Determining Dimensions of Thermoplastic Pipe and Fittings); ASTM D1243 (Intrinsic Viscosity of PVC)	DIN EN ISO 1163-1 (PVC Materials); DIN EN ISO 1628-2 (Viscosity Number of PVC Resins); DIN EN ISO 21306-1 (PVC Moulding Materials); DIN EN ISO 527 (Tensile Properties)	NSF/ANSI 61; NSF/ANSI 14; Chemical Process Industry Specifications
GPPS Resin	IS 10142 (Polystyrene Materials for Safe	ISO 1622-1 (Plastics — Polystyrene (PS) Moulding and Extrusion Materials — Designation System	ASTM D1892 (Polystyrene Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics);	DIN EN ISO 1622-1 (Polystyrene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow	FDA 21 CFR 177 (Food Contact Applications); UL 94 (where applicable); Plastics Industry Resin

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	and Basis for Specifications)	ASTM D1505 (Density of Plastics); ASTM D638 (Tensile Properties of Plastics)	Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	Specifications
HIPS Resin	IS 10142 (Polystyrene Materials for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 1622-1 (Plastics — Polystyrene (PS) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D1892 (Polystyrene Molding and Extrusion Materials); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 2897-1 (High Impact Polystyrene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; UL 94; Packaging Industry Guidelines
EPS Resin	IS 4671 (Expanded Polystyrene Products for Thermal Insulation Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 4899 (Expanded Polystyrene (EPS) — Specification)	ASTM D6817 (Rigid Cellular Polystyrene Foam), ASTM C578 (Rigid, Cellular Polystyrene Thermal Insulation), ASTM D1622 (Apparent Density of Rigid Cellular Plastics), ASTM D2126 (Response of Rigid Cellular Plastics to Thermal and Humid Aging)	DIN EN 13163 (Thermal Insulation Products Made of Expanded Polystyrene); DIN EN ISO 845 (Cellular Plastics — Apparent Density); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 12086 (Foam Materials, where applicable)	UL 94; FM Approvals (Insulation Applications); Building Code Fire Requirements
SAN Resin	IS 10146 (Styrene Acrylonitrile (SAN) Polymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 4894-1 (Plastics — Styrene/Acrylonitrile Copolymer (SAN) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4603 (Determination of Inherent Viscosity, where applicable); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 4894-1 (Styrene-Acrylonitrile Copolymers — Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; UL 94; Engineering Plastics Industry Specifications
ABS Resin	IS 17077(ABS molding and extrusion materials); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 2580-1 (Plastics — Acrylonitrile-butadiene-styrene (ABS) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4673 (Acrylonitrile-Butadiene-Styrene (ABS) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 2580-1 (ABS Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746A (Short-Term Material Evaluation); UL 746C (End Product Applications)
PET Resin (Bottle Grade)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use	ISO 18763 (Plastics — Poly(ethylene terephthalate) (PET) Moulding and Extrusion Materials — Designation System and Basis for	ASTM D4603 (Determination of Inherent Viscosity of Poly(Ethylene Terephthalate)); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM	DIN EN ISO 1628-5 (Determination of Viscosity Number of Polyethylene Terephthalate); DIN EN ISO 1133-1 (Melt Mass Flow Rate of	FDA 21 CFR 177; NSF/ANSI 51; Association of Plastic Recyclers (APR) Guidelines

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 14543 (Packaged Drinking Water Bottles); IS 7328; IS 10146; IS 10910; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	Specifications)	D1505 (Density of Plastics); ASTM D3418 (Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by DSC)	Thermoplastics); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 527 (Tensile Properties of Plastics)	
PET Resin (Film Grade)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 14636 (Flexible Packaging for Food Products); IS 7328; IS 10146; IS 10910; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 18763 (Plastics — Poly(ethylene terephthalate) (PET) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4603 (Determination of Inherent Viscosity of PET); ASTM D3418 (Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics)	DIN EN ISO 1628-5 (Viscosity Number of PET); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 11357 (Thermal Properties of Plastics by DSC); DIN EN ISO 1183-1 (Density of Plastics)	FDA 21 CFR 177; Flexible Packaging Association Guidelines; Film Industry Specifications
PET Resin (Fiber Grade)	IS 12252 (Polyethylene Terephthalate (PET)); IS 17264 (Polyester Fibre Applications); IS 7328; IS 10146; IS 10910; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 18763 (Plastics — Poly(ethylene terephthalate) (PET) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4603 (Intrinsic Viscosity of PET); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D3418 (DSC Thermal Properties of Polymers); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 1628-5 (Viscosity Number of PET); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 11357 (Thermal Properties by DSC)	OEKO-TEX Standard 100 (Textile Applications); Textile Exchange Standards (Recycled/Fibre Claims, where applicable)
PBT Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 7792-1 (Plastics — Thermoplastic Polyester (TP) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D7291 (Polybutylene Terephthalate (PBT) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 7792-1 (Thermoplastic Polyester Moulding Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	UL 94; UL 746A; Electrical and Electronics Industry Material Requirements
PC Resin	IS 14434 (Polycarbonate Roofing & Cladding Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS	ISO 21305-1 (Plastics — Polycarbonate (PC) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D3935 (Polycarbonate (PC) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 21305-1 (Polycarbonate Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Charpy Impact Strength)	UL 94; UL 746C; FDA 21 CFR 177 (Food Contact Grades)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	13360				
PMMA Resin	IS 10139 (Acrylic Resins for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 8257-1 (Plastics — Poly(methyl methacrylate) (PMMA) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D788 (Acrylic Plastic Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D1003 (Haze and Luminous Transmittance of Transparent Plastics)	DIN EN ISO 8257-1 (PMMA Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	UL 94; UL 746C; Building Material Fire Classification Requirements
PA6 Resin	IS 13360;IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530	ISO 1874-1 (Plastics — Polyamide (PA) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4066 (Nylon Injection and Extrusion Materials); ASTM D789 (Determination of Relative Viscosity of Polyamide (PA)); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 16396-1 (Polyamide Moulding and Extrusion Materials); DIN EN ISO 307 (Polyamides — Determination of Viscosity Number); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; FDA 21 CFR 177; Automotive Industry Material Specifications
PA66 Resin	IS 13360;IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530	ISO 1874-1 (Plastics — Polyamide (PA) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4066 (Nylon Injection and Extrusion Materials); ASTM D789 (Relative Viscosity of Polyamide); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 16396-1 (Polyamide Moulding and Extrusion Materials); DIN EN ISO 307 (Viscosity Number of Polyamides); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	UL 94; UL 746A; Automotive and Electrical Industry Material Specifications
POM Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 9988-1 (Plastics — Polyoxymethylene (POM) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D6778 (Polyoxymethylene (Acetal) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 9988-1 (Polyoxymethylene Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; NSF/ANSI 61; UL 94
PTFE Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 12086-1 (Plastics — Fluoropolymer Dispersions and Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4894 (PTFE Granular Molding and Ram Extrusion Materials); ASTM D4895 (PTFE Resin Produced from Aqueous Dispersion); ASTM D638 (Tensile Properties of Plastics); ASTM D1708 (Tensile Properties of Plastics by Microtensile Specimens)	DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN EN ISO 13000-1 (PTFE Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 177; USP Class VI; UL 94
PVDF Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS	ISO 12086-1 (Plastics — Fluoropolymer Dispersions and Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D3222 (Unmodified Poly(Vinylidene Fluoride) (PVDF) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 177; NSF/ANSI 61; Semiconductor Industry Material Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360				
UHMWPE Resin	IS 10910 (Polyethylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 7328; IS 10146; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 11542-1 (Plastics — Ultra-high Molecular Mass Polyethylene (PE-UHMW) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4020 (Ultra-High-Molecular-Weight Polyethylene Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D1505 (Density of Plastics); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 11542-1 (Ultra-High Molecular Weight Polyethylene Moulding Materials); DIN EN ISO 1183-1 (Density); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; USP Class VI; Medical Device Material Requirements
PEEK Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 16354 (Plastics — Polyetheretherketone (PEEK) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D6262 (Polyetheretherketone (PEEK) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 21318-1 (Polyetheretherketone Moulding and Extrusion Materials); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 177; USP Class VI; UL 94
PPS Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 20558-1 (Plastics — Polyphenylene Sulfide (PPS) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D4067 (Reinforced and Filled Poly(Phenylene Sulfide) (PPS) Molding and Extrusion Materials); ASTM D1238 (Melt Flow Rates of Thermoplastics); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load)	DIN EN ISO 1043-1 (Plastics — Symbols and Abbreviations); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746A; Electrical/Electronic Industry Material Specifications
PSU Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 21304-1 (Plastics — Polyarylsulfone (PSU) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D6394 (Polyphenylene Ether (PPE) Molding and Extrusion Materials, where applicable); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 21305-1 (Polycarbonate Materials, where applicable); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; USP Class VI; UL 94
PEI Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS	ISO 21304-2 (Plastics — Polyetherimide (PEI) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D5205 (Polyetherimide (PEI) Molding and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load); ASTM D256 (Izod Impact	DIN EN ISO 16396-1 (Polyamide Materials, where applicable); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746A; Aerospace/Electronics Material Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360		Resistance of Plastics)		
TPU Resin	IS 9845 (Overall Migration of Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 16365-1 (Plastics — Thermoplastic Polyurethane (TPU) Moulding and Extrusion Materials — Designation System and Basis for Specifications)	ASTM D789 (Polyamide Relative Viscosity, where applicable); ASTM D412 (Vulcanized Rubber and Thermoplastic Elastomers — Tensile Properties); ASTM D2240 (Rubber Property — Durometer Hardness); ASTM D624 (Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers)	DIN EN ISO 18064 (Thermoplastic Elastomers — Nomenclature and Classification); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 868 (Hardness of Plastics and Ebonite by Durometer); DIN EN ISO 34-1 (Tear Strength of Rubber and Thermoplastic Elastomers)	FDA 21 CFR 177 (Food Contact Grades); UL 94; Medical Device Material Requirements
EVA Resin	IS 10142 (Ethylene Vinyl Acetate for Food Contact Applications); IS 14636 (Flexible Packaging for Food Products) (Packaging Applications); IS 7328; IS 10146; IS 10910; IS 12252; IS 10151; IS 11434; IS 14534; IS 2530; IS 13360	ISO 21304-1 (Plastics — Ethylene-vinyl Acetate (EVA) Copolymer Materials — Designation System and Basis for Specifications, where applicable)	ASTM D3350 (Polyethylene Plastics Materials, where applicable); ASTM D1238 (Melt Flow Rate	DIN EN ISO 1873-1 (Polypropylene Materials, where applicable); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1183-1 (Density of Plastics)	FDA 21 CFR 177; Flexible Packaging Association Guidelines; Footwear Industry Specifications

Source field: Plastic and Plastic Products